Learning-Augmented Influence Maximization with Audited Marginal-Gain Oracles

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Abstract

Influence maximization selects a small set of seed nodes to maximize information diffusion in a network, but high-quality sequential selection requires repeatedly estimating conditional marginal gains with expensive influence oracles. Learned predictors can provide fast marginal-gain estimates, yet treating them as replacements for classical oracles is brittle: prediction errors are state dependent and can accumulate over the seed-selection trajectory. We study a learning-augmented alternative in which learned marginal gains are used as fallible advice rather than trusted decisions. Our framework first performs cheap full-graph screening, then combines a state-aware marginal proposal model with residual auditing, progressive verification, and a classical fallback. The resulting solver adaptively spends less oracle computation when learned advice is informative and automatically increases verification when the predictor becomes unreliable. Preliminary full-graph experiments support a favorable quality–efficiency tradeoff while controlled predictor corruption triggers the intended fallback behavior. These results suggest that reliable learning for influence maximization should focus not only on prediction accuracy, but on when and how predictions are verified inside the sequential optimization loop.

1 Introduction

Influence maximization (IM) is a canonical sequential optimization problem on social and information networks: given a diffusion graph and a seed budget, the goal is to select a small set of nodes whose activation maximizes expected influence spread. The classical greedy view is conceptually simple, but each greedy step can require comparing conditional marginal gains for many candidates, and accurately estimating these gains through Monte Carlo simulation or scalable sampling-based influence oracles remains computationally expensive on large graphs.

Learning offers an appealing shortcut. A graph model can predict which candidates are promising, or directly estimate the marginal gain of adding a candidate under the current seed set. However, a sequential IM solver needs more than an accurate predictor on average. The relevant ranking changes after every selected seed because of overlap and diminishing returns; a candidate that looks strong globally can become redundant under the current state. Consequently, a locally plausible prediction error can remove the true best candidate from consideration and propagate through later decisions. This makes a fully learned replacement for the influence oracle difficult to trust under distribution shift or predictor failure.

We take a different perspective: a learned marginal predictor should be treated as fallible advice to a classical optimizer, not as an oracle that must always be correct. The key question is therefore not “Can a GNN replace influence estimation?”, but rather “When is learned advice useful enough to reduce oracle work, and how much verification is needed before acting on it?” We instantiate this perspective with a learning-augmented IM framework that combines full-graph candidate screening, state-aware marginal proposals, an audited residual trust test, progressive candidate and sample verification, and a classical fallback (Figure 1). When predictions are useful, the learned model concentrates expensive evaluation on a small head of plausible candidates. When predictions are unreliable, the audit expands real-oracle computation and can recover classical behavior.

Our current evidence supports three central claims. First, marginal prediction must be explicitly state aware: supervision that exposes the same candidate under different seed states substantially improves conditional ranking sensitivity. Second, prediction alone is insufficient for reliable seed selection, whereas selective verification recovers much of the lost decision quality. Third, an audited progressive solver yields the desired learning-augmented behavior: useful advice reduces expensive verification, while corrupted or random advice increases oracle effort and activates fallback rather than causing catastrophic downstream degradation. Importantly, this behavior persists when the method receives the full influence graph and uses an independent screening stage to construct a compact candidate set.

Contributions. Our contributions are threefold:

- We formulate learned marginal-gain prediction for sequential IM as a fallible advice interface, highlighting why ordinary offline prediction quality is insufficient to guarantee reliable downstream decisions.
- We propose a learning-augmented solver that integrates full-graph screening, state-aware marginal proposals, residual auditing, progressive verification, and fallback to classical influence evaluation.
- We evaluate the resulting quality–efficiency–robustness tradeoff, including controlled predictor degradation and full-graph inputs, and show that oracle computation adapts to advice quality rather than assuming a uniformly reliable learned model.

2 Related Work

Classical influence maximization and scalable influence estimation. We will position the method against greedy influence maximization, Monte Carlo evaluation, and modern reverse-influence-sampling (RIS) solvers. The final version will emphasize strong paper-grade RIS baselines rather than weak heuristics.

Learning-based influence maximization. Prior GNN- and reinforcement-learning-based approaches learn node scores, seed-selection policies, or value/marginal surrogates. Our distinction is not that marginal gains have never been predicted; it is the explicit online interface that audits learned advice, allocates classical verification adaptively, and falls back when advice is unreliable.

Learning-augmented algorithms and prediction with verification. Our framework is most closely related in spirit to learning-augmented optimization: predictions are used to improve efficiency when helpful while a non-learned mechanism limits damage from poor advice. The final literature review will sharpen the connection to consistency/robustness and safe-verification principles without overstating guarantees that are not proved here.

3 Learning-Augmented Influence Maximization

3.1 Problem and Learned-Advice Interface

Let $G = (V, E)$ denote a diffusion graph and let $\sigma(S)$ be the expected spread of seed set $S \subseteq V$. Given a seed budget k, influence maximization seeks

$$\max_{S \subseteq V,\, |S| \le k} \sigma(S). \tag{1}$$

At a sequential state S, the conditional marginal gain of candidate $v \notin S$ is

$$\Delta(v \mid S) = \sigma(S \cup \{v\}) - \sigma(S). \tag{2}$$

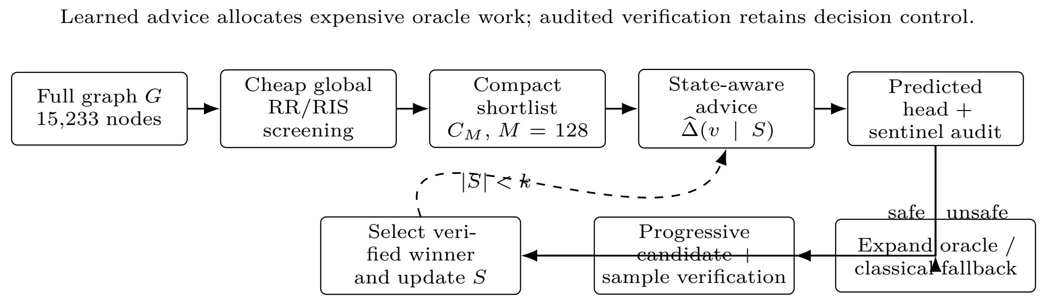

Figure 1: Overview of the learning-augmented IM pipeline. Independent full-graph screening creates a high-recall shortlist; a state-aware predictor proposes promising candidates; residual auditing decides whether progressive verification is sufficient or classical fallback is required.

We assume access to a learned advice model $\widehat{\Delta}(v \mid S)$ that is inexpensive relative to a high-fidelity classical influence oracle. Crucially, we make no assumption that this advice is uniformly accurate. The learned model proposes where expensive computation should be spent; an audited classical layer retains control of the actual seed decision. Figure 1 summarizes this interface.

3.2 Full-Graph Candidate Screening

The full graph is first reduced to a high-recall shortlist C_M using a cheap global RR/RIS sketch. This separates global scalability from the harder sequential conditional-ranking problem. Screening is deliberately independent of the learned marginal model: its role is only to retain plausible strong seeds before the state-dependent stage. Unless otherwise specified, our default shortlist size is $M = 128$, while $M \in \{64, 128, 256\}$ is used to expose the quality–cost sensitivity.

3.3 State-Aware Marginal Proposal

For candidates in $C_M \setminus S$, a graph predictor estimates $\widehat{\Delta}(v \mid S)$. Training explicitly exposes candidate nodes under multiple seed states and supervises conditional changes in gain. This matters because a predictor can otherwise learn a candidate-strength shortcut: it assigns nearly static scores to intrinsically strong nodes while missing overlap-induced diminishing returns. We therefore treat cross-state sensitivity of the same candidate as a first-class diagnostic rather than relying only on aggregate regression accuracy.

3.4 Audited Residual Trust Test

At each decision step, we evaluate a small predicted head together with rank-spaced sentinel candidates using a limited-budget classical oracle. For each audited candidate, define the residual

$$r(v \mid S) = \Delta_{\text{audit}}(v \mid S) - \widehat{\Delta}(v \mid S). \tag{3}$$

The audited residuals estimate whether an unverified outsider can plausibly overturn the current audited winner. The trust statistic is therefore aligned with the downstream failure event that matters: excluding a candidate that could beat the verified head. When the audit is inconclusive, the method spends more oracle work rather than accepting the learned top candidate by default.

3.5 Progressive Verification and Fallback

When the audit indicates that the current head is safe, the solver progressively increases candidate coverage and influence-estimation budget only as needed. If the residual audit signals unreliable advice, verification expands; in the limiting case the method falls back

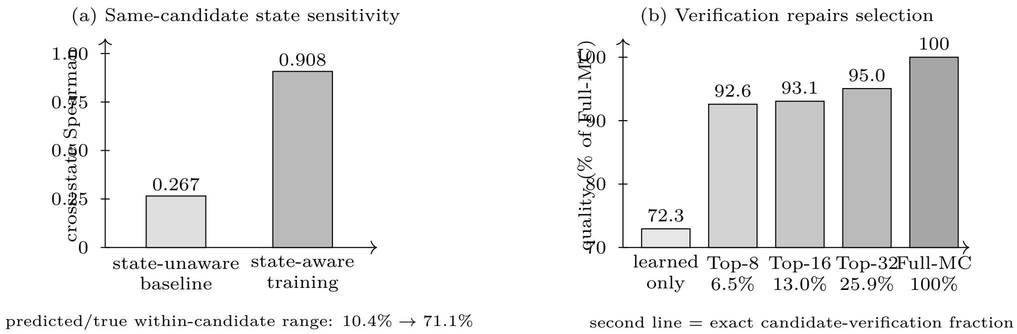

Figure 2: Diagnostics motivating the design. (a) State-aware supervision raises same-candidate cross-state Spearman from 0.267 to 0.908 and restores the response to diminishing returns. (b) Learned-only sequential selection reaches only about 72% of Full-MC quality; selectively verifying the predicted head recovers 92.6–95.0% as verification increases.

to all-candidate classical evaluation. Consequently, informative advice can reduce expensive evaluations while sufficiently poor advice drives computation back toward the classical solver.

Algorithm 1: Audited learning-augmented influence maximization.
Input: graph G, budget k, shortlist size M, learned advice $\widehat{\Delta}$, classical influence oracle.
1. Screen. Build a high-recall shortlist C_M from the full graph using an independent RR/RIS sketch.
2. Propose. At state S, score $v \in C_M \setminus S$ with $\widehat{\Delta}(v \mid S)$ and form a predicted head plus rank-spaced sentinels.
3. Audit. Evaluate the audit set with a small classical budget and estimate residual uncertainty relative to the current audited winner.
4. Verify progressively. If outsider risk is controlled, increase candidate coverage and sample budget only as needed to verify the next decision.
5. Fall back when needed. If the audit remains unsafe, expand classical evaluation up to the full candidate set.
6. Update. Add the verified winner to S and repeat Steps 2–5 until $|S| = k$.

4 Analysis

The theoretical goal of this paper is deliberately modest and aligned with what the algorithm actually guarantees. We will formalize (i) advice error, (ii) audit coverage, and (iii) verification cost. A conditional correctness proposition will show that, whenever the audited upper envelope validly covers the residual of every unverified outsider, the selected candidate matches the decision that would be obtained by the corresponding classical oracle. This yields a deterministic efficiency envelope: informative advice can reduce the number of expensive evaluations, whereas untrusted advice increases verification and the worst case recovers classical computation. We do not claim an unconditional approximation guarantee from the learned predictor itself.

5 Experiments

We organize the experiments around five research questions. Unless otherwise stated, the current full-graph closure uses NetHEPT under the independent-cascade model with $|V| = 15{,}233$, seed budget $k = 10$, an independent 50k-RR screening sketch, shortlist size $M = 128$, and 1,000 Monte Carlo worlds for final spread evaluation. Reported closure numbers use three independent repeats.

RQ1: Is learned marginal gain genuinely state aware? Figure 2(a) evaluates the same candidate under multiple overlap-controlled seed states. The state-unaware baseline attains

Table 1: Current full-graph NetHEPT closure ($k = 10$, $M = 128$, three repeats). "MC fraction" is expensive candidate-sample work relative to screened Full-MC selection; the independent RR reference uses a different RIS oracle and is therefore not assigned an MC fraction.

Method	Final spread	Quality vs. RR	MC fraction
Independent RR reference (50k)	506.74 ± 3.53	1.000	–
Screened Full-MC	508.47 ± 1.96	1.003 ± 0.011	1.000
Audited progressive (ours)	500.94 ± 2.44	0.989 ± 0.011	0.591 ± 0.085

mean same-candidate Spearman 0.267 and explains only 10.4% of the true within-candidate dynamic range. With state-aware supervision, these values improve to 0.908 and 71.1%, respectively; pairwise cross-state accuracy rises from 62.5% to 93.8%. This confirms that strong aggregate node ranking alone is insufficient for the conditional decisions required by sequential IM.

RQ2: Can learned advice reduce oracle work without sacrificing influence quality? Figure 2(b) first isolates the need for verification: learned-only selection compounds ranking errors, whereas fixed Top-m verification repairs much of the loss. We then test the full audited progressive solver from complete graph inputs. Table 1 shows that the proposed method retains $98.86\% \pm 1.09\%$ of the independent RR reference quality while using $59.08\% \pm 8.46\%$ of the expensive Monte Carlo candidate-sample work required by screened Full-MC selection.

RQ3: Does the solver degrade gracefully when predictions become unreliable? We evaluate clean, moderately corrupted, severely corrupted, and random advice while keeping the classical oracle unchanged. The desired learning-augmented behavior is not that quality remains independent of prediction quality; rather, decreasing trust should trigger increased real-oracle work before a bad proposal can propagate through the sequence. Controlled-corruption experiments support this mechanism: progressive verification without a residual audit can fail under random rankings, whereas the audited variant expands verification and activates fallback. The final common-protocol sweep reports quality and oracle work jointly.

RQ4: Does the method remain valid from full-graph inputs? The full NetHEPT graph is screened before any learned sequential decision. Across three independent screening seeds, the Top-128 RR shortlist contains all seeds selected by an independent RR reference (recall 1.00 ± 0.00). The resulting end-to-end quality in Table 1 therefore does not rely on constructing an artificial 128-node problem in advance; the compact decision set is produced from the complete graph by an independent global stage.

RQ5: Is the tradeoff stable across shortlist sizes, seed budgets, and graphs? For $M \in \{64, 128, 256\}$ on NetHEPT, independent-RR seed recall remains 1.00 in all three repeats. The audited solver exhibits a smooth quality–work tradeoff rather than a sharp dependence on a single shortlist size, motivating $M = 128$ as the default. The paper-grade evaluation expands this protocol across seed budgets and additional standard IM graphs, together with strong RIS baselines and a compact ablation of state-aware advice, residual auditing, progressive verification, and fallback.

Reproducibility Statement

We will provide the complete algorithmic specification, training protocol, dataset preprocessing, influence-model parameters, evaluation budgets, random-seed protocol, and all ablation settings in the main paper and appendix. Anonymous code and configuration files will be included as supplementary material so that the reported screening, marginal-prediction, verification, corruption, and fallback experiments can be reproduced from the same pipeline.

AI Use Statement

Generative AI tools were used during this project to assist with research planning and feedback on experimental design, software-code editing, literature search and summarization, interpretation and organization of experimental results, and drafting and editing parts of the manuscript. AI-assisted outputs are reviewed against source materials, executable experiments, and the authors' own analysis before inclusion. The authors take responsibility for the final content of the paper, including all claims, citations, code, experiments, and text produced with the assistance of generative AI. This disclosure is a working draft and will be reconciled with the complete research workflow before submission.

References

Bibliography placeholder for Draft v0.1. References will be populated from the verified GRL literature library.

A Additional Experimental Details

This appendix will contain full hyperparameters, dataset statistics, detailed corruption protocols, additional M and k sensitivity, per-seed results, runtime decomposition, and extended ablations.

B Additional Analysis

Complete proofs and any finite-sample audit discussion will be placed here once the corresponding statements are frozen.